Practice workbook · Finding the unknown letters in a rational curve

Working backwards from given asymptotes and intercepts to find the unknown letters in y = (ax + b)/(cx + d).

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-GRA-07

1. Question 3 · part b · 2 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part b only.

3

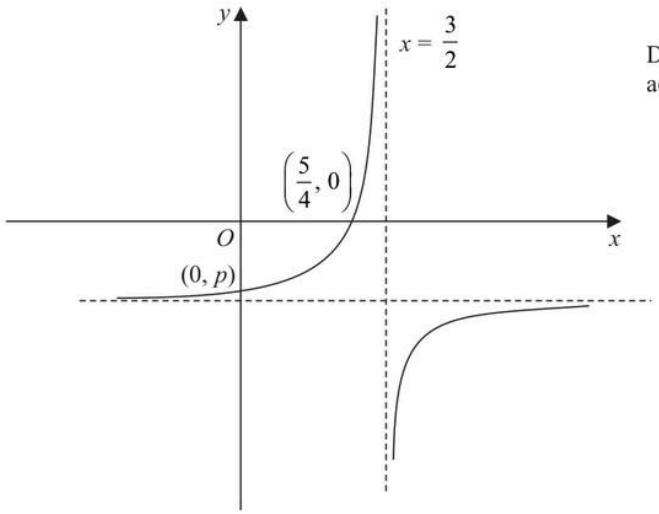


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sketch of the curve with equation

$$y = \frac{a - bx}{cx - d} \qquad x \neq \frac{d}{c}$$

where a, c and d are prime numbers and b is an integer.

The asymptote to the curve that is parallel to the y-axis has equation $x = \frac{3}{2}$

(a) Write down the value of c and the value of d

(2)

The curve crosses the x-axis at the point $\left(\frac{5}{4}, 0\right)$

(b) Find the value of a and the value of b

(2)

The curve crosses the y-axis at the point $(0, p)$ where p is a rational number.

(c) Find the value of p

(2)

(d) Find an equation of the asymptote to the curve that is parallel to the x-axis.

(1)

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8

P 6 6 0 2 7 R R A 0 8 3 6

1. Question 3 · part b · 2 marks · Working and answer

Working and answer

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2. Question 3 · part a · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

3 Curve C has equation $y = \dfrac{ax + 3}{1 - 2x}$ where $x \neq \dfrac{1}{2}$ and a is a constant.

The asymptote to C that is parallel to the x-axis has equation $y = 4$

(a) Find the value of a

(2)

(b) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

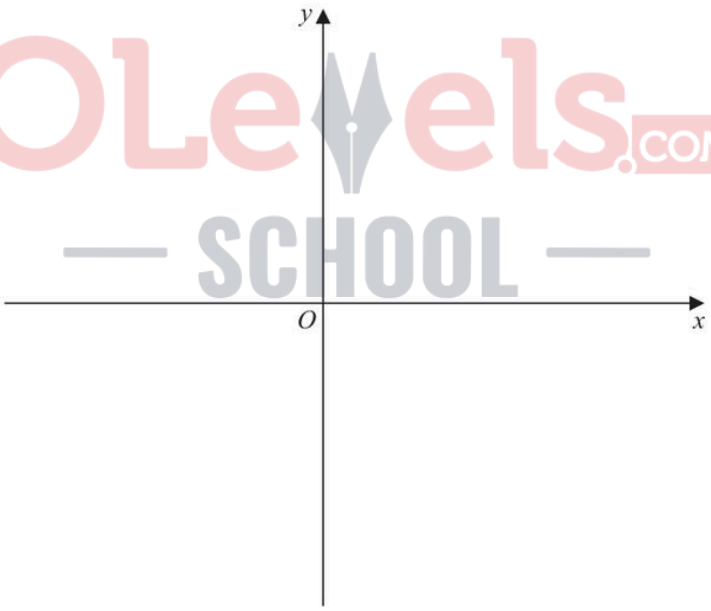
(c) Find the coordinates of the point where C crosses

(i) the x-axis, (ii) the y-axis.

(2)

(d) Using the axes below, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(4)



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2. Question 3 · part a · 2 marks · Working and answer

Working and answer

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3. Question 2 · part whole · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

2

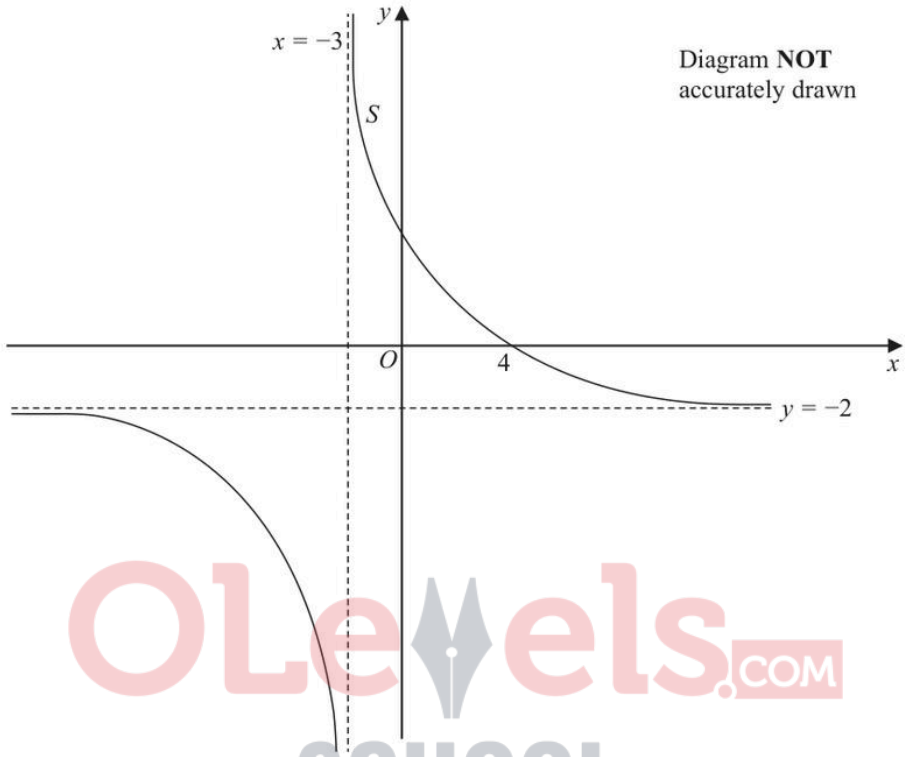


Figure 1

Figure 1 shows part of the curve S with equation $y = \dfrac{ax + b}{x + c}$ where a, b and c are integers.

The asymptote to S that is parallel to the x-axis has equation $y = -2$
The asymptote to S that is parallel to the y-axis has equation $x = -3$
The curve crosses the x-axis at the point with coordinates $(4, 0)$
The curve crosses the y-axis at the point with coordinates $(0, p)$ where p is a rational number.

Find

(i) the value of a,

(ii) the value of b,

(iii) the value of c,

(iv) the value of p.

(4)

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P 6 1 8 8 3 A 0 4 3 6

3. Question 2 · part whole · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 4 marks)

3. Question 2 · part whole · 4 marks · Working and answer

Working and answer

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4. Question 4 · part a.i · 0 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

4

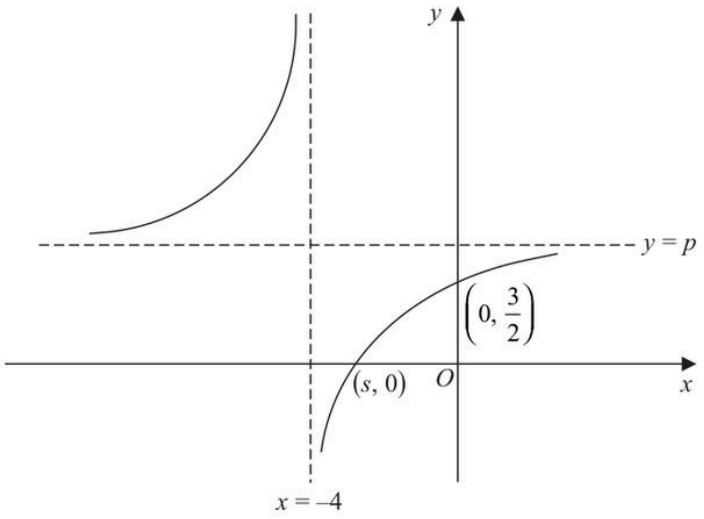


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve C with equation

$$y = \frac{2x + q}{x + r} \qquad x \neq -r$$

where q and r are integers.

The asymptote to C that is parallel to the y-axis has equation $x = -4$
The asymptote to C that is parallel to the x-axis has equation $y = p$

(a) Write down

(i) the value of p

(ii) the value of r

(2)

Given that C crosses the y-axis at the point with coordinates $\left(0, \frac{3}{2}\right)$

(b) find the value of q

(2)

Given that C crosses the x-axis at the point with coordinates $(s, 0)$

(c) find the value of s

(2)

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4. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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5. Question 4 · part a.ii · 2 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

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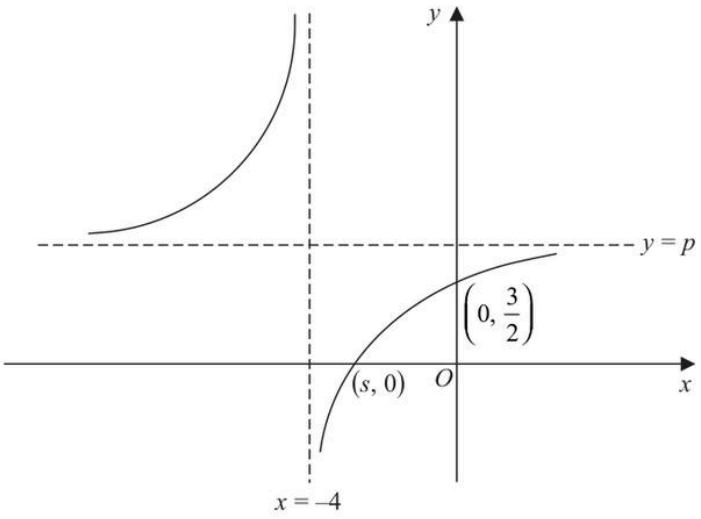


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Figure 1

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where q and r are integers.

The asymptote to C that is parallel to the y-axis has equation $x = -4$
The asymptote to C that is parallel to the x-axis has equation $y = p$

(a) Write down

(i) the value of p

(ii) the value of r

(2)

Given that C crosses the y-axis at the point with coordinates $\left(0, \frac{3}{2}\right)$

(b) find the value of q

(2)

Given that C crosses the x-axis at the point with coordinates $(s, 0)$

(c) find the value of s

(2)

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5. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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6. Question 4 · part b · 2 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

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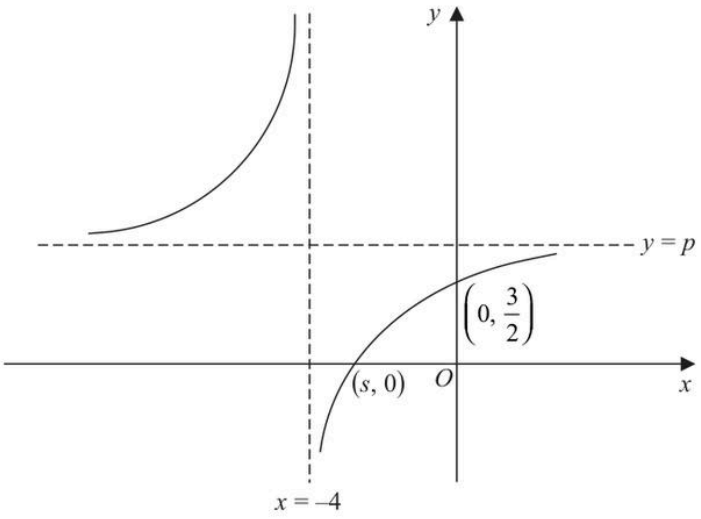


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Figure 1

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$$y = \frac{2x + q}{x + r} \qquad x \neq -r$$

where q and r are integers.

The asymptote to C that is parallel to the y-axis has equation $x = -4$
The asymptote to C that is parallel to the x-axis has equation $y = p$

(a) Write down

(i) the value of p

(ii) the value of r

(2)

Given that C crosses the y-axis at the point with coordinates $\left(0, \frac{3}{2}\right)$

(b) find the value of q

(2)

Given that C crosses the x-axis at the point with coordinates $(s, 0)$

(c) find the value of s

(2)

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6. Question 4 · part b · 2 marks · Working and answer

Working and answer

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7. Question 10 · part a.i · 0 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

(ii) Find the value of a

(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

(c) Find the coordinates of the point where C crosses the x-axis.

(1)

(d) Using calculus, show that at every point on C, the gradient is positive.

(4)

(e) Using the axes on the next page, sketch C
Clearly label the asymptotes and the coordinates where C crosses the coordinate axes.

(3)

The gradient of C is $\frac{13}{100}$ at the point E and at the point F

(f) Find the length of the line EF, giving your answer to 2 decimal places.

(6)

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7. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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8. Question 10 · part a.ii · 3 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

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(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

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8. Question 10 · part a.ii · 3 marks · Working and answer

Working and answer

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9. Question 10 · part a · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part a only.

10 The curve C has equation $y = \dfrac{ax - 5}{b - x}$ where a and b are integers and $x \neq b$

One intersection of C with the coordinate axes is at the point with coordinates $\left(\frac{5}{4}, 0\right)$

The asymptote parallel to the y-axis has equation $x = 3$

(a) Find the value of a and the value of b

(2)

(b) Sketch C, showing clearly the asymptotes with their equations and the coordinates of the points of intersection with the coordinate axes.

(5)

The straight line l with equation $4y - 7x = k$ has no points of intersection with C

(c) Show, using algebra, that the range of possible values of k can be written as

$$m < k < n$$

where m and n are integers to be found.

(9)

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9. Question 10 · part a · 2 marks · Working and answer

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10. Question 3 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

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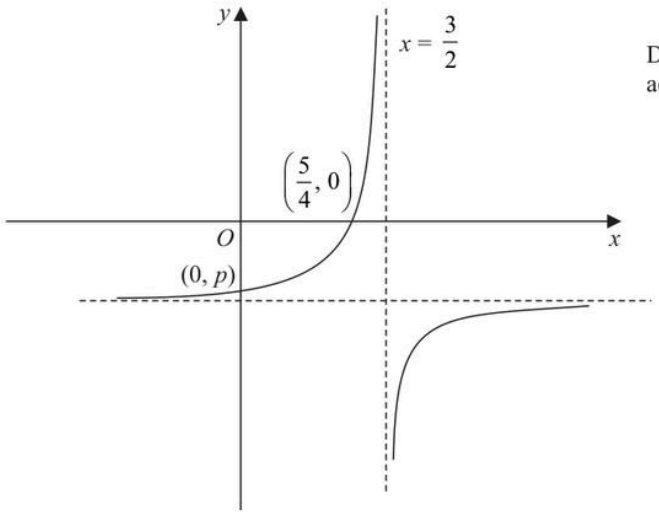


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where a, c and d are prime numbers and b is an integer.

The asymptote to the curve that is parallel to the y-axis has equation $x = \frac{3}{2}$

(a) Write down the value of c and the value of d

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(b) Find the value of a and the value of b

(2)

The curve crosses the y-axis at the point $(0, p)$ where p is a rational number.

(c) Find the value of p

(2)

(d) Find an equation of the asymptote to the curve that is parallel to the x-axis.

(1)

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10. Question 3 · part a · 2 marks · Working and answer

Working and answer

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11. Question 11 · part b.ii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part b.ii only.

11 A curve C has equation

$$y = \frac{(2a-1)x+1}{ax-6} \qquad \text{where } a \text{ is a constant and } x \neq \frac{6}{a}$$

(a) Find $\frac{dy}{dx}$

(3)

The curve crosses the y-axis at the point A.

The normal to C at the point A is the line l with equation $66y - 72x + 11 = 0$

Show that

(b) (i) $a = 3$

(4)

(ii) the equation of C is $y = \frac{5x+1}{3x-6}$ where $x \neq 2$

(1)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes with their equations and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(d) Find the x coordinate of D.
Give your answer as an improper fraction.

(4)

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11. Question 11 · part b.ii · 1 mark · Working and answer

Working and answer

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